

PIEZOELECTRICS

Biodegradable ferroelectric molecular crystal with large piezoelectric response

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Transient implantable piezoelectric materials are desirable for biosensing, drug delivery, tissue regeneration, and antimicrobial and tumor therapy. For use in the human body, they must show flexibility, biocompatibility, and biodegradability. These requirements are challenging for conventional inorganic piezoelectric oxides and piezoelectric polymers. We discovered high piezoelectricity in a molecular crystal $\text{HOCH}_2(\text{CF}_2)_3\text{CH}_2\text{OH}$ [2,2,3,3,4,4-hexafluoropentane-1,5-diol (HFPD)] with a large piezoelectric coefficient d_{33} of ~ 138 picocoulombs per newton and piezoelectric voltage constant g_{33} of $\sim 2450 \times 10^{-3}$ volt-meters per newton under no poling conditions, which also exhibits good biocompatibility toward biological cells and desirable biodegradation and biosafety in physiological environments. HFPD can be composite with polyvinyl alcohol to form flexible piezoelectric films with a d_{33} of 34.3 picocoulombs per newton. Our material demonstrates the ability for molecular crystals to have attractive piezoelectric properties and should be of interest for applications in transient implantable electromechanical devices.

The piezoelectric effect is a phenomenon in which certain dielectrics generate electronic polarization through an applied stress, or conversely (1–3). The effect enables direct conversion between electrical and mechanical energy, which allows materials to be widely used in actuators, transducers, sensors, and energy harvesters (4–7). The piezoelectric effect can exist in ionic, atomic, and molecular crystals (8–10). Among them, piezoelectric ionic crystals, represented by ferroelectric oxides barium titanate (BTO) and lead zirconate titanate (PZT), have a wide range of industrial and commercial applications because of their excellent electromechanical conversion, high Curie temperature, and strong spontaneous polarization as well as their being usable in the polycrystalline ceramic form (11–13). In comparison, piezoelectric molecular crystals generally have weak piezoelectric performance caused by relatively weak intermolecular forces, with piezoelectric coefficient $d_{33} < 40$ pC/N (table S1) (5). Ferroelectric polymers, such as polyvinylidene fluoride (PVDF), belong to piezoelectric molecular semicrystals, which have been widely used in flexible electronic devices on the basis of their high pliability, light weight, and low toxicity (14–19).

Transient implantable piezoelectric materials are attracting interest for biosensing, drug delivery, tissue regeneration, and antimicrobial and tumor therapy because nondegradable scaffolds and implanted devices would increase the risk of infection and inflammation (20–22). For example, Liu *et al.* reported a biodegradable poly(L-lactic acid) piezoelectric nanofiber scaffold that can be used as a battery-less electrical stimulator to promote chondrogenesis and cartilage regeneration (21). Piezoelectrics intended for human body application should be biocompatible, biodegradable, and flexible (23–26). However, many of these requirements cannot be met with conventional inorganic ferroelectric oxides and ferroelectric polymers, especially those pertaining to biodegradability. Piezoelectric molecular crystals, which are basically composed of organic single-component compounds, may be a promising solution to these concerns because of their decent solubility, potential biocompatibility, and biodegradability (27). For example, biomolecular crystals can provide many potentially beneficial properties of biomaterials such as flexibility, biocompatibility, reliability, biodegradability, and environmental sustainability (28–31). Yang *et al.* successfully created wafer-scale heterostructured piezoelectric biomaterial thin films based on γ -glycine crystals (22). However, those piezoelectric biomaterials have very weak piezoelectric performance, with piezoelectric coefficients of $d_{33} \leq 10$ pC/N, such as lysozyme (6.5 pC/N) (29), DL-alanine (4.8 pC/N) (30), collagen (0.89 pC/N) (31), and γ -glycine (10.4 pC/N) (22, 28), all of which are much smaller than that of PVDF (~ 28 pC/N) (table S1) (32).

We report a piezoelectric molecular crystal, $\text{HOCH}_2(\text{CF}_2)_3\text{CH}_2\text{OH}$ [2,2,3,3,4,4-hexafluoropentane-1,5-diol (HFPD)], that shows strong

piezoelectric response, with a large d_{33} reaching ~ 138 pC/N and a piezoelectric voltage constant g_{33} of $\sim 2450 \times 10^{-3}$ Vm/N under no poling conditions. Discovery of a molecular crystal with a d_{33} of more than 100 pC/N is a notable development since the discovery in 1880 of the piezoelectric effect (table S1) (5, 22, 33). The HFPD–polyvinyl alcohol (PVA) films not only exhibit excellent flexibility, high biocompatibility, and no obvious toxicity to biological cells but also show great biodegradation and biosafety in physiological environments. Our HFPD–PVA (2:1) films demonstrate excellent piezoelectricity, having a d_{33} of 34.3 pC/N, which is much larger than that of other organic piezoelectric composites formed by biofriendly molecular crystals such as γ -glycine (5.3 pC/N) (22). A two-dimensional (2D) hydrogen bond network through O–H...O interactions of terminal hydroxyl in molecules induces decent solubility in various solvents and results in the biodegradability of HFPD crystal. The anisotropy of Young's modulus comes from the ordered arrangement of a 2D hydrogen bond network and fluorine atoms arranged on the chain in the crystal structure, which might be an important reason for the large piezoelectric coefficient of HFPD crystal.

Crystal structures and phase transition behaviors

We recrystallized transparent rod-like single crystals of HFPD with the size of 1 by 1 by 5 mm using the solvent diffusion method at room temperature (Fig. 1A) (34). These crystals showed good environmental stability (fig. S1) and thermal stability up to 380 K (fig. S2). Single-crystal x-ray diffraction (XRD) determined that HFPD adopts polar crystal symmetry with $P1$ space group at room temperature (table S2). HFPD molecules connected with the adjacent ones (Fig. 1B) to form a 2D hydrogen bond layer through O–H...O hydrogen bonding interactions (Fig. 1C and table S3). The asymmetric distribution of the C–F bonds generates a non-zero molecular dipole moment, which is superimposed on each other by the 2D hydrogen bond layered stacking structure, resulting in the generation of spontaneous polarization (fig. S3). When the crystal was cooled below 259 K, we observed slight changes in cell parameters that revealed the occurrence of isomorphous phase transitions (Fig. 1B and table S2).

Differential scanning calorimetry (DSC) curves and temperature-dependent dielectric measurements confirmed that HFPD shows a thermal anomaly at 282 K upon heating (Fig. 1D). Temperature-dependent XRD (fig. S4) and Raman spectrum measurements (Fig. 1E) further verified this thermodynamic phase transition. We also found that this structural phase transition can be triggered under pressure. To confirm this, we performed in situ pressure-dependent Raman spectrum measurements using

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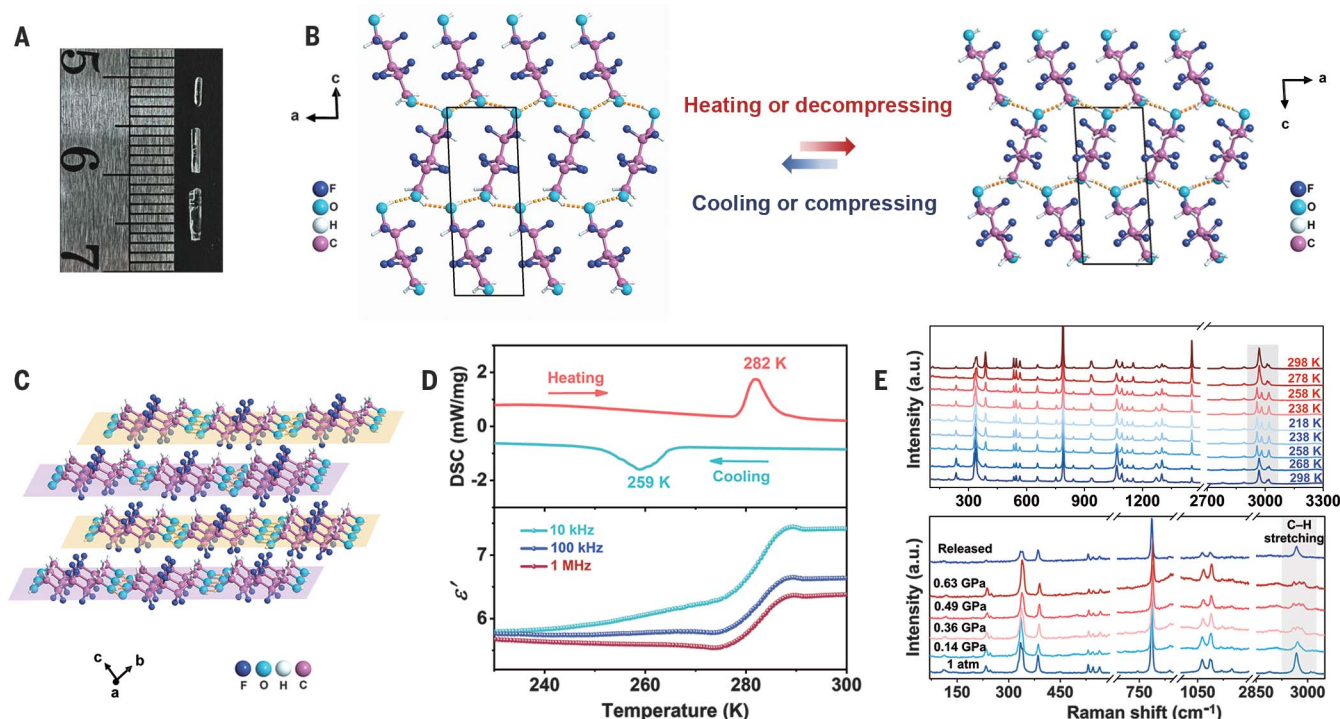


Fig. 1. Crystal structures and phase transitions of HFPD. (A) Digital image of HFPD crystals. (B) The transition of 2D monolayer formed through O–H...O interactions of terminal hydroxyl in molecules between two phases triggered by thermal or stress fields. The yellow dashed lines indicate the O–H...O hydrogen bonding interactions. (C) Packing view of

2D hydrogen bond layers. (D) The results of DSC and dielectric (along the *b* axis) measurements, revealing a structural phase transition at around 282 K. (E) (Top) Temperature-dependent and (bottom) pressure-dependent Raman spectra show thermal- and pressure-induced phase transitions.

diamond anvil cells. The pressure-dependent Raman spectra exhibited a peak transformation around 2970 cm^{-1} (C–H stretching) when applied with pressure up to 0.36 GPa, whereas a strong narrow peak separated into three successive peaks at 2958, 2972, and 2983 cm^{-1} (Fig. 1E). The triple-peak character returned to initial single-peak form after the pressure was released from 0.63 GPa, indicating the full reversibility of the pressure-induced phase transition. The Raman vibrations recorded at the low-temperature phase correspond to those of the high pressure-induced new phase, indicating that these two phases should be the same. This was also confirmed with single-crystal XRD of HFPD crystal under a high pressure of 0.6 GPa (table S4).

Ferroelectric properties

HFPD crystals have the desired ferroelectricity with switchable spontaneous polarization because their polarization directions are roughly perpendicular to the long axis of the molecules, where the rotational reorientation of molecules along the long axis can relatively easily achieve polarization reversal. We implemented polarization–electric field (*P-E*) hysteresis loop tests for the HFPD to confirm their ferroelectricity, using the double-wave method. The HFPD crystals exhibit well-shaped *P-E* hysteresis loops

along different crystal orientations (Fig. 2, A to C). The HFPD crystals hold moderate saturated polarization (P_s) of 0.73, 4.88, and $6.34\text{ }\mu\text{C}/\text{cm}^2$ along the *a*, *b*, and *c* axes, respectively, indicating obvious crystal anisotropy, which is also reflected in the different coercive field values. The coercive field is substantially lower than that of PVDF, with a typical value of hundreds of kilovolts per centimeter (18). We measured the breakdown electric field of HFPD to be $726.74\text{ kV}/\text{mm}$, which is higher than that of PVDF ($346.57\text{ kV}/\text{mm}$) (fig. S5 and table S5).

We further investigated the ferroelectric behaviors and piezoresponse of HFPD on the nanoscale by using piezoresponse force microscopy (PFM) characterization, which is a powerful technique with which to characterize ferroelectrics by measuring the dynamic electromechanical response (35–38). The PFM phase image clearly shows a polydomain structure of the HFPD crystalline thin film, indicating the existence of spontaneous polarization in different directions (Fig. 2D and fig. S6). To confirm the ferroelectricity of HFPD, we obtained the box-in-box domain pattern (Fig. 2E and fig. S7) by successively applying a tip voltage of +90 V and –80 V to a region with an initial single-domain state. This reversible domain-switching behavior further verifies the ferroelectricity of HFPD. Moreover, we performed

the PFM switching spectroscopy measurement on the crystalline thin-film surface. The butterfly amplitude loops and 180° phase shifts when the amplitude signal was at a minimum, demonstrating robust ferroelectric switching (Fig. 2F).

Piezoelectric performance

We used the PFM measurements on the HFPD crystalline thin films to evaluate their piezoelectric performance. The XRD pattern indicated that the orientation of HFPD crystal in its crystalline thin film is mainly at the (012) and (010) planes (fig. S8). We show in Fig. 3A the relationship between piezoelectric amplitude and frequency of the HFPD and PVDF when the excitation frequency sweeps around the contact resonance frequency. We fit the curves with the damped harmonic oscillator model, and HFPD exhibits a much larger piezoresponse peak (34). We can obtain the intrinsic inverse piezoelectric deformation of the sample by acquiring the inverse optical lever sensitivity of the PFM cantilever and correcting the resonance magnification using *Q* factor. We repeated such measurements over six randomly chosen points on the HFPD crystalline thin film and PVDF film under different ac excitation voltages, respectively. It shows that the electromechanical responses from HFPD and PVDF adapt linear relationships with respect to their

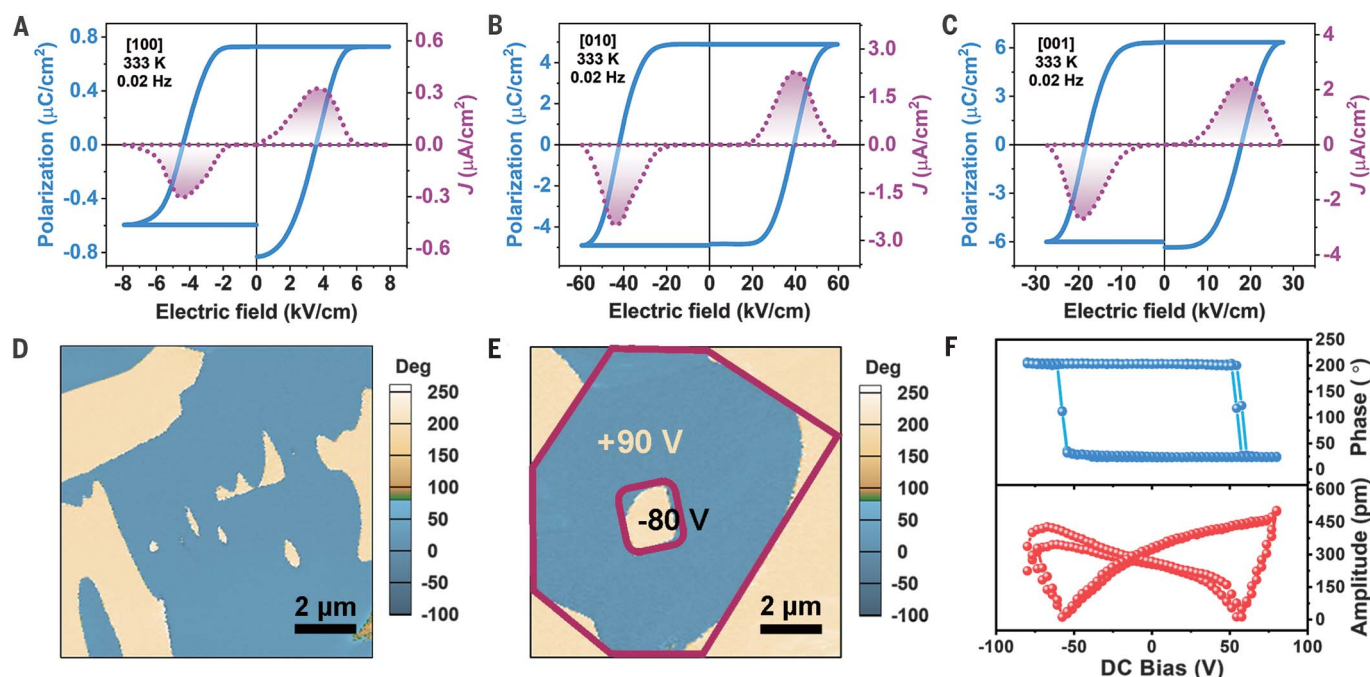


Fig. 2. Ferroelectric properties of HFPD. (A to C) *P-E* hysteresis loops for HFPD single crystals along (A) [100], (B) [010], and (C) [001] crystal orientations, respectively. (D) PFM phase image captured on HFPD crystalline thin film. (E) PFM phase image showing a box-in-box domain

pattern, which was obtained by successively applying a tip voltage of +90 V and −80 V to a region with an initial single-domain state. (F) PFM switching hysteresis loops for amplitude and phase signals on HFPD crystalline thin film.

piezoelectricity (Fig. 3B). We can estimate the piezoelectric coefficient from the slope, which we measured to be 119.3 pm/V for HFPD crystalline thin films and 18.8 pm/V for PVDF film (21 pC/N measured with the quasi-static method).

We investigated the piezoelectric response of HFPD crystals with the Berlincourt method, based on the direct piezoelectric effect. We measured the longitudinal piezoelectric coefficients d_{33} of HFPD crystals to be ~90 pC/N on the (010) plane, ~70 pC/N on the (012) plane, and a maximum of ~138 pC/N on the (021) plane (Fig. 3C and fig. S9). Such promising d_{33} values of HFPD crystals are considerably large among single-component organic piezoelectrics and are competitive with the piezoelectric polymer PVDF (34). The piezoelectric voltage coefficient, as a key indicator of performance, directly represents a material figure of merit for piezoelectric sensors. We estimated the g_{33} of HFPD to be about 2450×10^{-3} Vm/N. This value is nine times higher than that of PVDF (122.4×10^{-3} to 263.5×10^{-3} Vm/N) and seven times that of poly(vinylidene fluoride-*ran*-trifluoroethylene) [P(VDF-TrFE)] (225.9×10^{-3} to 357.7×10^{-3} Vm/N) (table S6) (32). Also, we calculated the electrostriction coefficient Q_{22} of HFPD crystal to be about $15.28 \text{ m}^4/\text{C}^2$, which is larger than that of PVDF ($1.3 \text{ m}^4/\text{C}^2$) (34).

To investigate the mechanical properties of our crystals, we calculated the elastic stiffness matrix c_{ij} (34). The small elastic stiffness con-

stant suggests that HFPD crystal is an inherently mechanically soft material. To understand the elastic properties more intuitively, we plotted the directional Young's modulus, which represents the uniaxial stiffness in different directions (Fig. 3D). The directional Young's modulus has a very large anisotropy, and the two maxima are located in an irregular plane that is almost elliptical. The anisotropy of Young's modulus comes from the ordered arrangement of 2D hydrogen bond network and F atoms arranged on the chain in the crystal structure. In addition, the spatial dependence of shear modulus MIN surfaces with six maxima are regularly nested within the MAX surfaces to form a shell structure of shear modulus (Fig. 3, E and F). Such an exotic mechanical property might be an important reason for the large piezoelectric coefficient of the HFPD crystal. The pressure-induced structural phase transition may be another reason. It can be found that under a certain pressure, some of the HFPD molecules in the lattice are likely to rotate 45° , resulting in a certain increase in d_{33} (figs. S10 and S11 and movie S1) (34). To compare with the experimentally used piezoelectric strain constants (commonly denoted as d_{ij}), the piezoelectric stress coefficients e_{ij} and elastic compliances s_{ij} should be calculated because $d_{ij} = e_{ij} * s_{ij}$, in which the elastic compliance s_{ij} is obtained by inverting the elastic stiffness matrix c_{ij} directly. Combining the values of e_{ij} and c_{ij} matrix, we can directly get the piezoelectric strain con-

stants d_{ij} (table S7)

37.6	−101.8	64.2	−109.1	−455.3	−266.7
13.7	−46.4	29.6	−50.4	−166.2	−93.0
−8.0	26.1	−21.7	17.3	94.7	52.6

According to the calculation results of this matrix, the maximum allowable longitudinal piezoelectric coefficient is up to 248 pC/N, which means that the large value of d_{33} may be experimentally measured, providing competitive candidates for piezoelectric sensors and fine-tuned microelectromechanical systems. Furthermore, we used the Berry phase method to estimate the ferroelectric polarization of the crystal, from which the polarization with −0.93, 4.22, and 6.22 $\mu\text{C}/\text{cm}^2$ along the *a*, *b*, and *c* axes can be extracted, in good accordance with the symmetry requirement of space group *P1*. We estimated by means of vector summation the total polarization to be 7.58 $\mu\text{C}/\text{cm}^2$, which is close to the measured value.

Flexible piezoelectric films

For piezoelectric devices compatible with the human body such as artificial muscle, electronic skin, and wearable devices, their flexibility is of great concern. Current flexible piezoelectrics mainly include PVDF-based polymers, inorganic-polymer composites, and bio-piezoelectric materials (18, 22, 27, 39, 40). The HFPD crystals present an elastic modulus (*E*) of 6.14, 8.85, and 4.30 GPa and hardness (*H*) of 437.4, 365.1, and 507.3 MPa for (100), (010), and (001) planes,

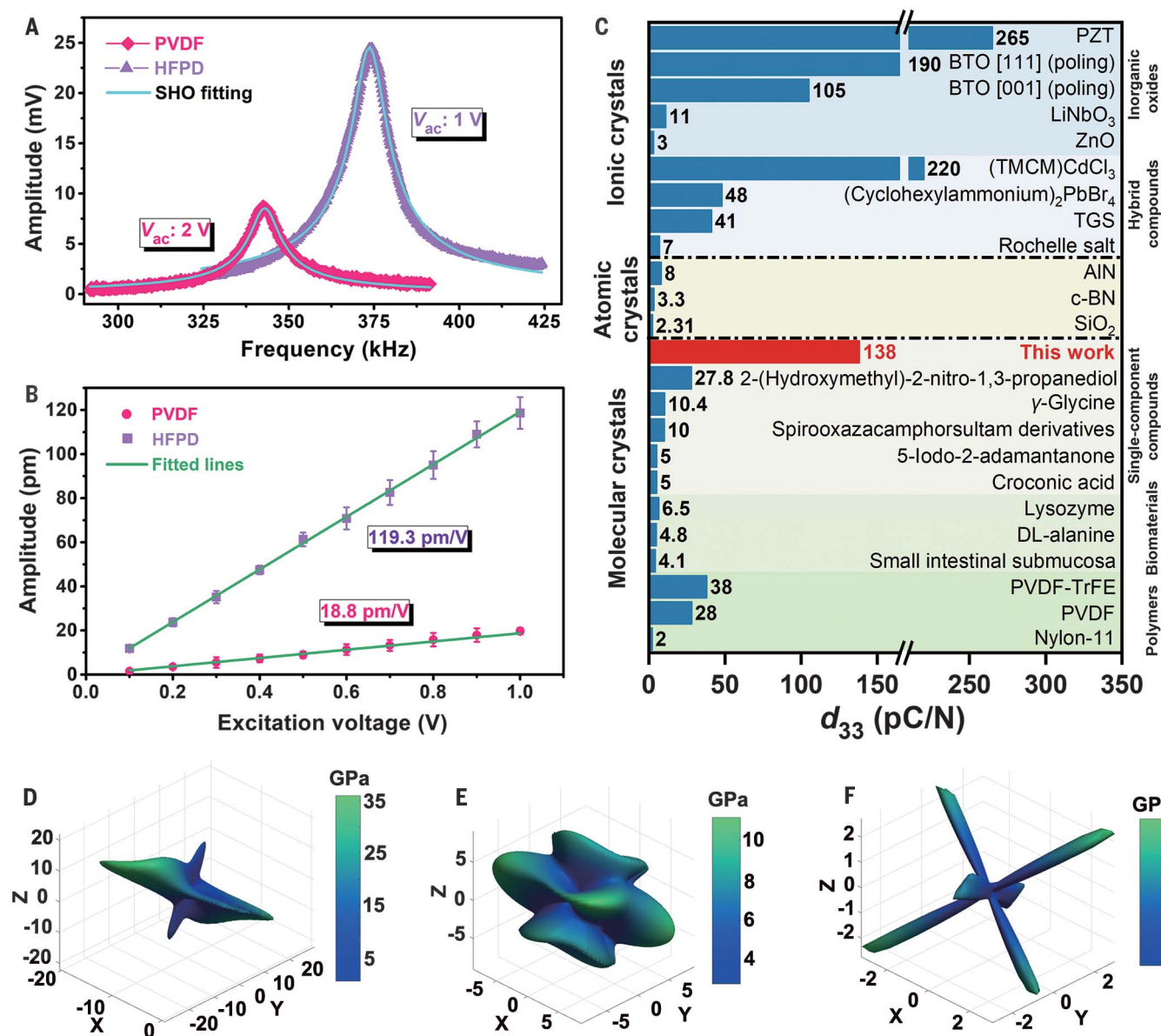


Fig. 3. Piezoelectric properties of HFPD. (A) Piezoresponse of HFPD and PVDF films versus excitation frequency measured with resonance-enhanced PFM and analyzed by the damped harmonic oscillator model. (B) The derived amplitude-excitation curve for HFPD and PVDF films by PFM tip. (C) Piezoelectric coefficient of HFPD compared with some classical piezoelectric ionic, atomic, and molecular crystals, including PZT (42), BTO (43), (TMCM)CdCl₃ [trimethylchloromethyl

ammonium trichlorocadmium(II)] (5), triglycine sulfate (TGS) (44), γ -glycine (45), AlN (46), lysozyme (29), PVDF (32), poly(vinylidene fluoride-co-trifluoroethylene) (PVDF-TrFE) (47), croconic acid (5), and more (the detailed data are included in table S1). Measured d_{33} values are listed at the end of each bar. (D to F) 3D plots of elastic modulus of HFPD crystals. (D) Directional Young's modulus. (E) Spatial dependence of shear modulus MAX. (F) Spatial dependence of shear modulus MIN.

respectively (fig. S12) (34), in the same magnitude as PVDF ($E = 3.28$ GPa, $H = 240.3$ MPa) (47), indicating the great application potential in flexible and body-compatible electromechanical devices. Furthermore, the acoustic impedance (z_0) for HFPD [4.08×10^6 kg/(m² s)] is lower than that of P(VDF-TrFE) [4.51×10^6 kg/(m² s)], which is comparable with that of PVDF [3.92×10^6 kg/(m² s)], which matches that of the body reasonably well, making it a promising candidate for implantable ultrasound applications (table S8) (32).

Crystals are inherently rigid and brittle, which makes them challenging to be used in flexible systems. We thus prepared the HFPD-PVA films by means of solution evaporation from their corresponding mixture solution at 40°C (Fig. 4A) (34). The films exhibited excellent flexibility, biocompatibility, and piezoelectricity (Fig. 4A). The as-prepared films with the HFPD-to-PVA ratio above 2:1 were relatively uniform (fig. S13). The top-view (Fig. 4B and fig. S14) and cross-sectional scanning electron microscope (SEM) (fig. S14) images show the dis-

tribution of HFPD crystal in the HFPD-PVA films. Among them, the corresponding elemental mapping image demonstrated that F elements (only from HFPD) were uniformly distributed in the HFPD-PVA (2:1) film (Fig. 4B and fig. S14). We measured stress-strain curves to understand the mechanical properties of as-prepared films with various HFPD-to-PVA ratios (Fig. 4C and table S9). The maximum tensile strains of these films decreased gradually with the increased amounts of incorporated HFPD, indicating the degradation of their stretchability (Fig.

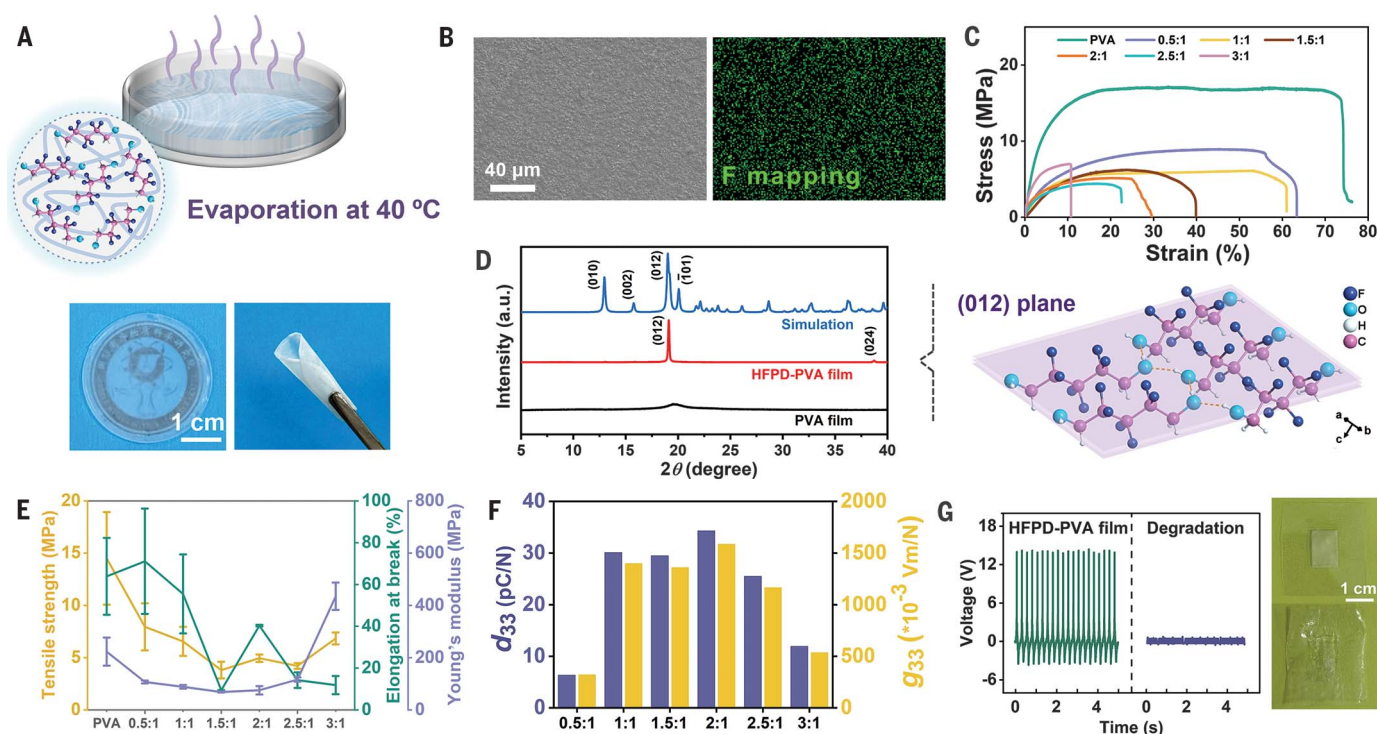


Fig. 4. Characterization of HFPD-PVA films. (A) (Top) Schematic diagram of the film fabrication by solution evaporation and digital images of the (bottom left) as-grown and (bottom right) curled film showing good flexibility. (B) (Left) SEM image of the as-prepared HFPD-PVA (2:1) film and (Right) corresponding elemental mapping for F element. (C) Stress-strain curves of as-prepared films with various HFPD-to-PVA ratios. (D) (Left) XRD patterns simulated from the single crystal of HFPD and measured from the as-prepared HFPD-PVA (2:1) and

pure PVA film. (Right) Schematic diagram of molecular arrangement in (012) plane. (E) Young's modulus, elongation at break and tensile strength of as-prepared films with various HFPD-to-PVA ratios. (F) The piezoelectric coefficient d_{33} and piezoelectric voltage coefficient g_{33} obtained from the as-prepared films with different HFPD-to-PVA ratios. (G) (Left) The piezoelectric voltage output of a packaged HFPD-PVA film with a 2:1 ratio before and after being immersed in PBS. (Right) Digital images of as-prepared film encapsulated with PLA before and after degradation.

4E). Neat PVA possessed a tensile strength of 14.5 ± 4.43 MPa, Young's modulus of 223.18 ± 53.88 MPa, and elongation at break of $63.92 \pm 18.37\%$. For films with a HFPD-to-PVA ratio less than 1.5:1, the crystalline samples were not distributed homogeneously on the films (fig. S13), which made their mechanical properties fluctuate within a certain range. When the HFPD-to-PVA ratio was greater than 1.5, we observed a substantial increase in Young's modulus. Hence, the films within the HFPD-to-PVA ratio of 1.5 to 2.5 demonstrate the most balanced stretchability and Young's modulus (Fig. 4E). The combination of PVA and HFPD piezoelectric crystals largely maintains the good mechanical flexibility of polymers but sacrifices the strength to some degree.

XRD spectra obtained from the as-prepared HFPD-PVA (2:1) films showed one prominent diffraction at 19.1° , which corresponds to the (012) plane of the HFPD single crystal (Fig. 4D). From the perspective of crystallography, the (012) plane of HFPD crystal is the 2D hydrogen bond layer, where HFPD molecules are connected with the adjacent ones through O-H...O hydrogen bonding interactions (Fig. 4D). Thus, it could be rationally suggested

that the continuous crystallization of HFPD with orientation in the as-prepared films could mainly account for the induction of piezoelectricity. We examined the piezoelectric performances of films made from different HFPD-to-PVA ratios by using the Berlincourt method (Fig. 4F). The highest piezoelectric coefficient d_{33} (~ 34.3 pC/N) was observed in the film with a 2:1 ratio. This value is comparable with that of the typical ferroelectric polymer PVDF and far beyond those of most reported bio-organic films such as DL-alanine (4.8 pC/N) (30), collagen (0.89 pC/N) (31), and γ -glycine (10.4 pC/N) (22). On the HFPD-PVA film with the ratio of 2:1, the measured d_{33} values were distributed from 32 to 37 pC/N (table S10), confirming the good uniformity of this piezoelectric film. Because of their low permittivity (table S11), the piezoelectric voltage coefficient g_{33} value of HFPD-PVA film with the ratio of 2:1 was calculated to be 1587.6×10^{-3} Vm/N. This value is far beyond that of PZT ceramics (20×10^{-3} to 40×10^{-3} Vm/N) and an order of magnitude larger than that of PVDF. Such a high g_{33} value indicates a large voltage output of the composite film, which is critical to the performance of piezoelectric sensors. Films

with other HFPD-to-PVA ratios also exhibited appreciable g_{33} values (Fig. 4F and table S11).

We further investigated the piezoelectric performance of the HFPD-PVA films by measuring the generated output voltages after repetitively striking them with a certain external mechanical stress. An impulse force of 40 N was repeatedly applied to the film surface at a frequency of 4 Hz. The as-prepared HFPD-PVA (2:1) films exhibited a stable output peak-to-peak voltage (V_{pp}) of around 18 V (Fig. 4G). We measured the force- and frequency-dependent piezoelectric output of the HFPD-PVA (2:1) films. With the force and frequency increasing, the generated output voltages gradually increased (fig. S15). Because both PVA and HFPD are water soluble, the HFPD-PVA films dissolved into an aqueous solution after about 1 hour (fig. S16). We thus used biodegradable polylactic acid (PLA) encapsulation to construct a controllable transient electromechanical device, in which the duration time for the device to work depends on the encapsulation properties. After being immersed in the phosphate buffer solution (PBS) for several days, the packaged composite film was degraded, and the electrical signal V_{pp} decreased to zero (Fig. 4G).

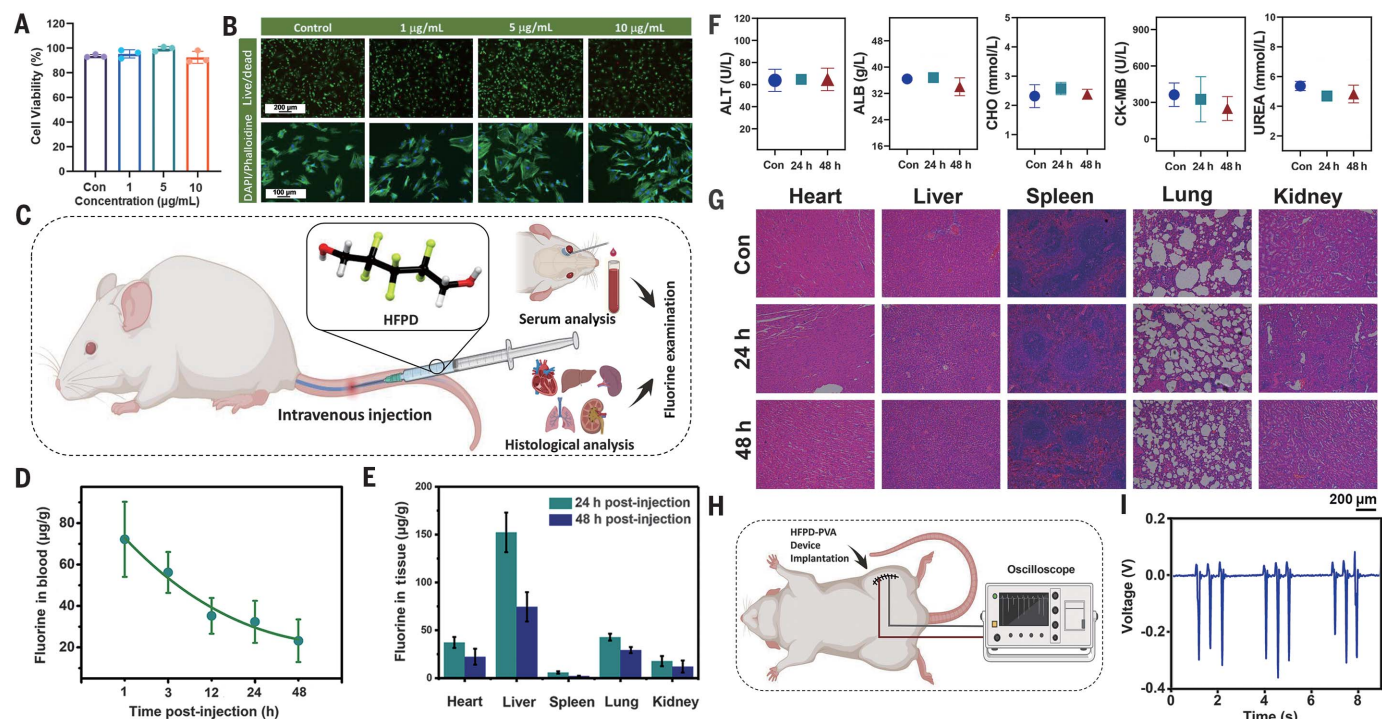


Fig. 5. Biocompatibility and biodegradability of HFPD. (A) Viability of rBMSCs after incubation with various concentrations of HFPD. (B) Representative images of live/dead staining (green and red labels indicate living and dead cells, respectively) and phalloidine/DAPI staining (green and blue labeled cells indicate cytoskeleton and nucleus of cells, respectively) of rBMSCs after incubation with various concentrations of HFPD. (C) Schematic diagram of the process of in vivo biocompatibility and biodegradability assessment through tail intravenous injection of HFPD. (D) Time-dependent changes in fluorine levels in the blood of rats after intravenous injection of HFPD (80 mg/kg) ($n = 6$ rats per group). (E) Distribution of fluorine levels in major organs of rats, including heart, liver,

spleen, lung, and kidney of rats at 24 and 48 hours after injection. The fluorine amounts were determined by using ion chromatography after tissue homogenization using hydrogen peroxide ($n = 6$ rats per group). (F) Results of serum biochemical tests conducted on treated rats ($n = 6$ rats per group). (G) Hematoxylin and eosin (H&E) staining images of major organs collected from control and rats administered with HFPD at 24 and 48 hours after injection. Control rats received PBS injections. (H) Schematic diagram of the piezoelectric performance measurement of PLA-packaged HFPD-PVA device in SD rat. (I) Representative voltage generation of the implanted PLA-packaged HFPD-PVA device in the rat knee area.

Biocompatibility and biodegradability

We confirmed the in vitro biocompatibility of this piezoelectric crystal by culturing it with multiple types of cells, such as rat bone marrow-derived mesenchymal stem cells (rBMSCs), mouse calvarial pre-osteoblast cell line MC3T3-E1 cells, and rat neural stem cells (rNSCs). We conducted cell viability tests in culture medium supplemented with 1, 5, and 10 µg/ml of HFPD using the Cell Counting Kit-8 (CCK-8) assay. As illustrated in Fig. 5A, HFPD demonstrated excellent biocompatibility toward rBMSCs, with cell viability exceeding 90% even at a concentration of up to 10 µg/ml. Furthermore, live/dead and phalloidine/DAPI (4',6-diamidino-2-phenylindole) staining were performed to evaluate the potential cytotoxicity of HFPD toward rBMSCs. As depicted in Fig. 5B, nearly all the cells treated with various concentrations of HFPD exhibited green fluorescence and similar spreading behavior accompanied by extended filopodia compared with that of the control group. Similar trends could be observed in MC3T3-E1 cells treated with various concentrations of HFPD (fig. S17). Furthermore, our organic fluo-

ride piezoelectric HFPD showed no neurotoxin, as evidenced by the high cell viability (fig. S18) and no impact on the differentiation ability of rNSCs (fig. S19). Taken together, these results undoubtedly indicated that HFPD was of great biocompatibility.

We conducted in vivo biodegradability assessment through tail intravenous injection of HFPD (80 mg/kg) into Sprague-Dawley (SD) rats for 24 and 48 hours, as schematically depicted in Fig. 5C. We measured the time-dependent examination of the fluorine amounts in blood (Fig. 5D); major organs including the heart, liver, spleen, lung, and kidney (Fig. 5E); as well as excreta, including urine and feces (fig. S20) using ion chromatography after tissue homogenization, and the results further substantiated that HFPD was biodegradable. Orbitrap high-resolution mass spectrometry of urine in SD rats intravenously injected with HFPD solution (80 mg/kg) confirmed that HFPD molecules could be removed from the body (fig. S21). Additionally, the analysis of serum parameters—including alanine aminotransferase (ALT), albumin (ALB), cholesterol (CHO), creatine

kinase-MB (CK-MB), and urea—all fell within the normal range (Fig. 5F). Histological examinations of major organs revealed no noticeable structural or cellular alterations (Fig. 5G).

Furthermore, the biosafety of HFPD was validated by implanting a HFPD-PVA film (10 by 10 mm) under the skin of SD rats. As expected, all the serum parameters were within the normal range (fig. S22). Neither major organs (fig. S23) nor the skin tissue directly above the HFPD-PVA film (fig. S24) exhibited discernible inflammatory cell infiltration or pathological features. To further extend the potential biomedical application of HFPD-PVA film, we constructed the PLA-packaged HFPD-PVA (2:1) device and tested its piezoelectric performance in vivo in SD rats. We implanted devices under the skin in the posterior limb for substantial mechanical-electrical transduction detection (Fig. 5H). When we briefly pulled and released the rat's right leg at a certain beat, the embedded device that was attached to the knee area produced the corresponding electrical signal V_{pp} of ~300 mV (Fig. 5I). All of these results demonstrated that HFPD had great

biodegradation and biosafety, and these superior performances suggest that HFPD-PVA films possessed great application potential as carriers for drug delivery, self-powered transient energy-harvesting devices for therapeutics, and biodegradable high-performance piezoelectric scaffolds for regenerative medicine.

Summary

We have presented a simple ferroelectric molecular crystal HFPD with piezoelectricity better than PVDF, and the obtained piezoelectricity does not require the pretreatments of stretching and poling. An intriguing 2D hydrogen bond network in HFPD shows distinct advantages for film-forming fabrication and pollution-free recycling. The HFPD-PVA films not only exhibit excellent flexibility, high biocompatibility, and no obvious toxicity to biological cells, they also show great biodegradation and biosafety in physiological environments. Our finding provides a fundamental template and potential inspiration for the development of transient implantable electromechanical devices.

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SUPPLEMENTARY MATERIALS

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Biodegradable ferroelectric molecular crystal with large piezoelectric response

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Editor's summary

Piezoelectric materials are useful for sensing strain or generating ultrasonic vibrations useful for biomedical applications. However, they are too often rigid and brittle and are not biodegradable. Zhang *et al.* found a molecular crystal with a high piezoelectric response that may be attractive for biomedical applications (see the Perspective by Wang and Li). The molecular crystals can be made into thin, strain-sensing films with good flexibility. The authors also describe promising observations for biocompatibility and biodegradability of the molecular crystal. —Brent Grocholski

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